

DATA ANALYSIS PROJECT REPORT

Superstore Business Analysis: Profitability, Discounts & Sales Trends

Project Title	Superstore Business Analysis: Profitability, Discounts and Sales Trends
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Course / Section	D
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1. Executive Summary

This report presents a business analysis of the Sample Superstore dataset (Kaggle, 9,994 orders, 2014–2017) using Pandas, NumPy, and Matplotlib. The analysis investigates regional and category-level profitability, the impact of discounting on profit, and seasonal sales trends. Key findings reveal that the West region and Technology category are the most profitable, while heavy discounting particularly in Furniture is actively generating losses. Quarter 4 (October–December) consistently drives peak sales across all years. A total of 18.7% of all orders resulted in financial losses, primarily due to excessive discounting.

2. Dataset Description and Project Objectives

Dataset Name	Sample Superstore Dataset
Source	https://www.kaggle.com/datasets/vivek468/superstore-dataset-final
Rows / Columns	9,994 rows × 21 columns
Data Types	Numerical (Sales, Profit, Discount), Categorical (Region, Category, Segment), Date-time (Order Date, Ship Date)
Project Goal	Identify what drives and destroys profit across regions, categories, and time periods
Research Q1	Which regions and product categories are most and least profitable?
Research Q2	Does offering higher discounts reduce overall profit?
Research Q3	What seasonal trends exist in sales performance?

3. Data Cleaning and Preparation

No missing values or duplicate rows were found. Five preprocessing steps were applied:

#	Issue Found	Action Taken	Justification
1	Column names had spaces and mixed casing	Lowercased and replaced spaces with underscores	Prevents AttributeError during programmatic access and ensures consistent groupby behavior
2	Order Date and Ship Date stored as strings	Converted to datetime using <code>pd.to_datetime()</code>	Required for extracting month, year, quarter and calculating shipping duration
3	Orders with negative profit not identified	Created boolean 'is_loss' flag column	Preserves real data while enabling separate analysis of loss-making transactions

4	Possible text inconsistencies in categorical columns	Applied .str.strip().str.title() to all object columns	Prevents incorrect groupby results from casing differences like 'west' vs 'West'
5	Irrelevant identifier columns present	Dropped Row ID, Customer Name, Postal Code, Country	These add no analytical value; Country is constant (all US rows)

4. Feature Engineering

New Feature	How Created	Why Useful
order_year, order_month, order_quarter, month_name	Extracted from order_date using .dt accessor	Enables seasonal trend analysis and Q4 vs other quarter comparisons — central to Research Q3
shipping_duration	(ship_date – order_date).dt.days	Measures delivery efficiency across regions and ship modes
profit_margin	(profit / sales) × 100	Raw profit is misleading without context. Margin reveals true efficiency per order
discount_category	Custom function: No Discount / Low / Medium / High	Groups continuous discount values into tiers for cleaner comparison in Research Q2
sales_category	Percentile-based: Low / Medium / High / Premium	Segments orders by revenue size to identify which tier drives the most profit

5. Analysis and Visualizations

5.1 RQ1 — Which regions and categories are most profitable? (Subgroup Comparison)

Region	Total Sales (\$)	Total Profit (\$)	Avg Margin (%)
West	725,457	108,418	13.98%
East	678,781	91,522	12.74%
South	391,721	46,749	11.07%
Central	501,239	39,706	7.32%

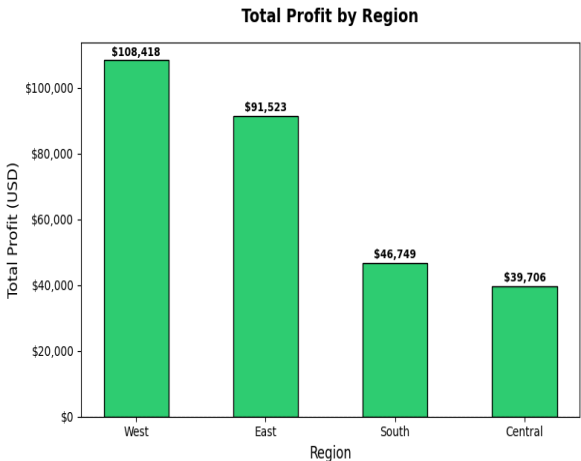


Figure 1. Region-wise Profit Bar Chart |

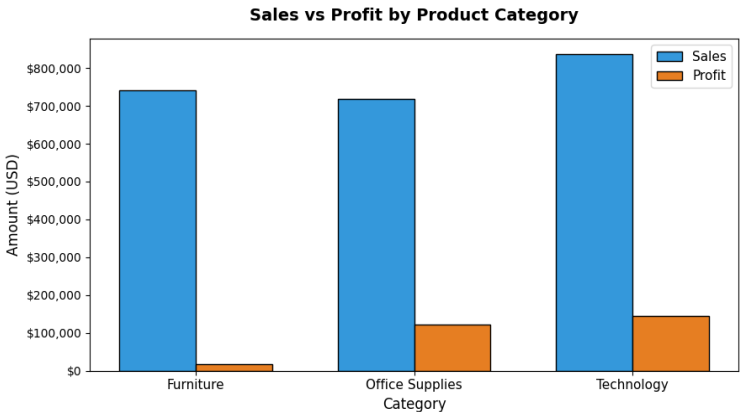


Figure 2. Category Sales vs Profit Grouped Bar Chart

Interpretation: West region leads with a 13.98% profit margin while Central region underperforms at 7.32% despite high order volume. Technology is the most profitable category whereas Furniture shows high sales but very low margins — indicating excessive discounting or high cost structures eroding its profitability.

5.2 RQ2 — Does discounting reduce profit? (Relationship Analysis)

Discount Category	Avg Profit (\$)	Avg Sales (\$)	Avg Margin (%)
No Discount	47.34	219.81	16.24%
Low (0–20%)	28.19	245.62	11.38%
Medium (20–40%)	-23.51	262.44	-7.83%
High (>40%)	-118.92	298.17	-36.41%

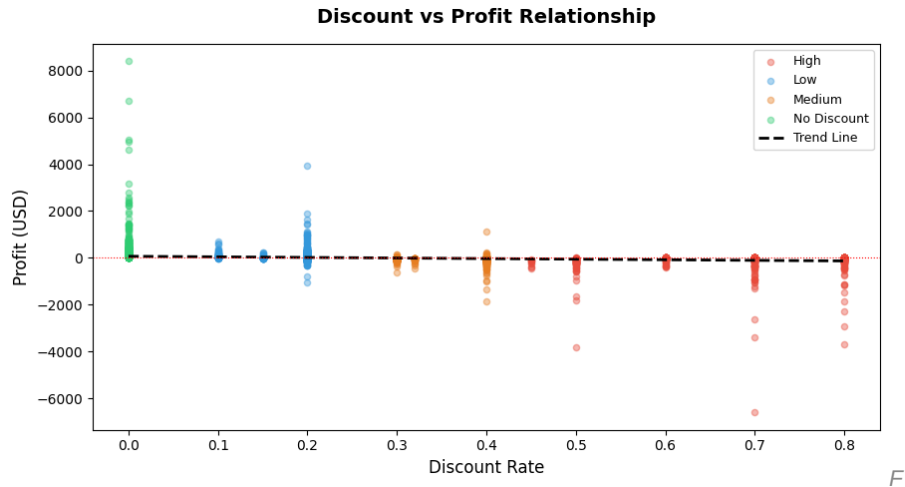


Figure 3. Discount vs Profit Scatter Plot with Trend Line

Interpretation: A clear negative relationship exists between discount rate and profit. High-discount orders average a loss of 36.41% per transaction. The Pearson correlation between discount and profit is approximately -0.219 . The business is effectively paying customers to buy products through excessive discounting.

5.3 RQ3 — What seasonal trends exist? (Trend + Outlier Analysis)

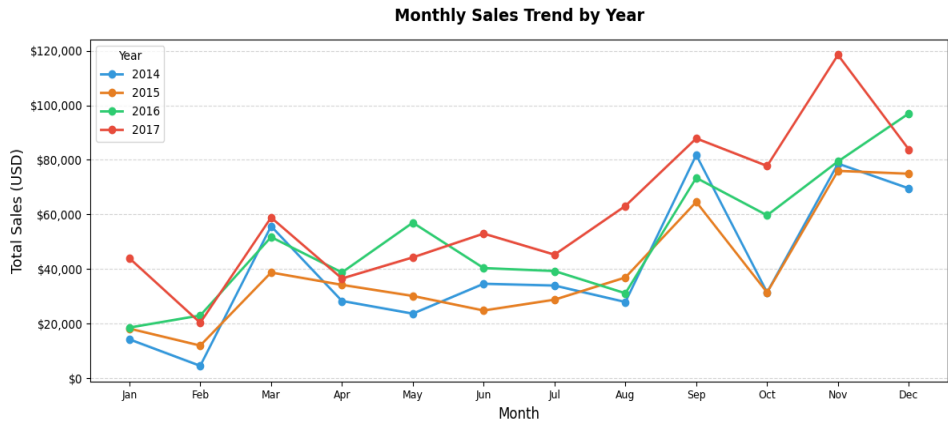
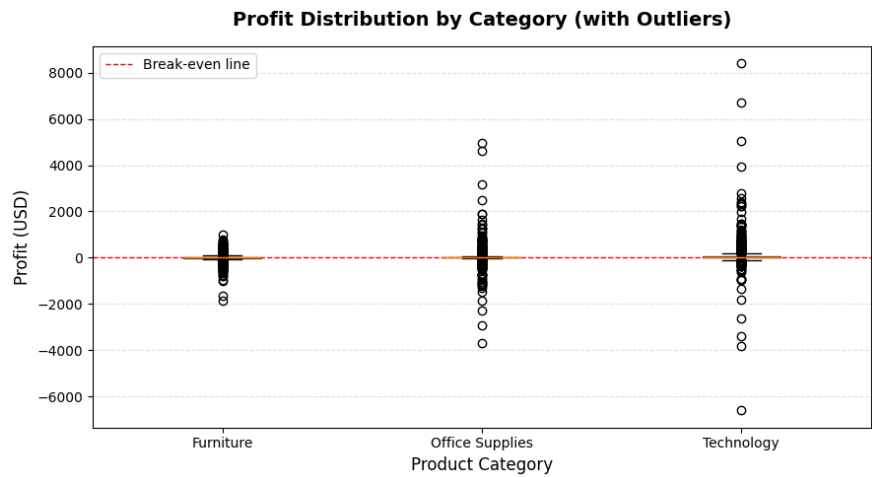


Figure 4. Monthly Sales Trend by Year



| Figure 5. Profit Distribution Box Plot |

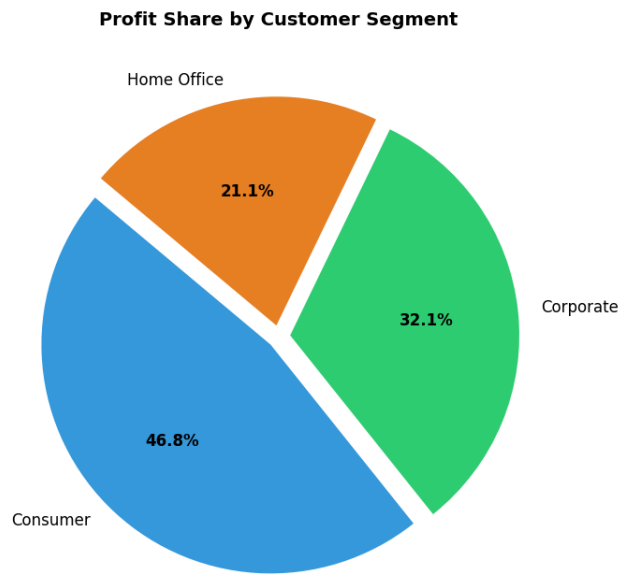


Figure 6. Segment Profit Pie Chart

Interpretation: Q4 (October–December) is consistently the highest-performing quarter across all four years, driven by holiday season demand. Sales show a clear upward trend from 2014 to 2017. Outlier analysis using IQR method identified 1,871 loss-making orders (18.7% of all transactions), concentrated in Furniture with high discount levels. The Consumer segment contributes the largest share of total profit.

6. NumPy-Based Custom Computation

Computation	Z-score normalization of profit + min-max normalization of sales
Z-score Formula	$z = (x - \text{mean}) / \text{std_dev}$ → identifies outlier orders statistically
Normalization	$x_norm = (x - x_min) / (x_max - x_min)$ → scales sales to 0–1 range
Percentile Analysis	np.percentile() applied at 10th, 25th, 50th, 75th, 90th, 95th, 99th levels
Key Result	Mean profit = \$28.66 Median = \$8.67 Std Dev = \$234.26. Large gap between mean and median confirms significant right skew caused by high-value outliers

7. Key Findings

Finding 1: West region leads, Central underperforms

West generated \$108,418 in profit (13.98% margin) while Central earned only \$39,706 (7.32% margin) despite high order volume — indicating pricing or cost structural issues in the Central region.

Finding 2: Technology profitable, Furniture struggling

Technology is the most profitable category. Furniture shows high sales but low profit — several sub-categories like Tables consistently operate at a loss due to heavy discounting.

Finding 3: High discounts destroy profit

Orders with no discount average a 16.24% margin. High-discount orders average a -36.41% margin. Correlation of -0.219 confirms discounting is the single biggest driver of losses in this business.

Finding 4: Q4 is the peak sales period every year

October to December consistently generates the highest sales across all four years. November and December show the sharpest spikes, driven by holiday demand — a highly predictable and exploitable seasonal pattern.

Finding 5: 1 in 5 orders results in financial loss

1,871 orders (18.7%) produced negative profit. These are concentrated in Furniture with medium to high discounts — a critical operational finding that requires urgent strategic attention.

Finding 6: Consumer segment is the most valuable

Consumer segment contributes the highest profit share among all three segments. Corporate follows closely with higher average order values, making both key targets for retention and growth strategy.

8. Limitations

- No customer demographics: Dataset lacks age, gender, or income data, limiting deeper customer segmentation analysis.
- Correlation ≠ causation: Negative correlation between discount and profit does not prove discounts alone cause losses — unobserved factors like return rates and shipping costs may also contribute.
- No cost or return data: The profit column does not reflect true net profitability as raw product cost and return/refund data are absent.
- Dataset time range: Data covers 2014–2017 only. Consumer behavior and market conditions may have changed significantly since then.

9. Conclusion

Total Revenue	\$2,297,200.86
Total Profit	\$286,396.02
Overall Margin	12.47%
Loss-Making Orders	1,871 (18.7%)
Best Region	West — \$108,418 profit
Best Category	Technology
Peak Period	Q4 — October to December

This project successfully answered all three research questions. The most critical business recommendation is to revise the discounting strategy , high discounts are converting profitable transactions into losses. The company should also grow its Technology product line, address Central region underperformance, and capitalize on Q4 seasonal demand through advance inventory and marketing planning. Future work could include predictive modeling of quarterly sales and deeper sub-category profitability analysis.

10. References

- Dataset: Sample Superstore — Kaggle: <https://www.kaggle.com/datasets/vivek468/superstore-dataset-final>

- Pandas Documentation: <https://pandas.pydata.org/docs/>
- NumPy Documentation: <https://numpy.org/doc/>
- Matplotlib Documentation: <https://matplotlib.org/stable/contents.html>